

MENOFYIA UNIVERSITY

SYSTEM ANALYSIS & DESIGN PROJECT

SECOND YEAR

2026-2027



جامعة المنوفية

كلية الحاسبات والمعلومات

2026-2027

## AirVoyage

| <i>No.</i> | <i>Name</i>                    | <i>Section</i> |
|------------|--------------------------------|----------------|
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| 2          | Mouamen Mohamed Kamel Elshazly | 1              |
| 3          | Adam Ehab Maher                | 1              |
| 4          | Ziad Osama Hassan Mostafa      | 1              |
| 5          | Yousef Mohamed Doma Rashed     | 1              |

# Project Identification

## Project Name:

Travel Flights Booking System

## Current Problems:

- 1- High Error Rate in Manual Data Entry
- 2- Operational Delays
- 3- Vulnerability to Data Loss and Damage
- 4- Wasting Time and Effort
- 5- Lack of Accuracy

## Description of project:

- The system aims to create a comprehensive qualitative shift by automating all daily operations and procedures instead of traditional and manual methods. The system relies on a secure, centralized digital database that ensures records and data are stored with high accuracy while preventing any possibility of damage or loss, along with providing periodic backup capabilities. Furthermore, the system provides organized and fast working environment that seamlessly connects different departments, significantly reducing human errors and speed up task execution with high efficiency. In addition, the system is characterized by its ability to generate instant, accurate, and constantly updated reports, giving management a comprehensive and clear vision to assist decision-makers in making precise and effective decisions based on reliable data.

# System Request

| <b><i>System Request - Travel Flights Booking System</i></b> |                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Sponsor</b>                                       | Mohamed                                                                                                                                                                                                                                         |
| <b>Business Need:</b>                                        | <ul style="list-style-type: none"><li>1- Flight Booking Automation</li><li>2- 24/7 Service Availability</li><li>3- Real-Time Flight and Seat Updates</li><li>4- Enhanced Customer Experience</li><li>5- Decision-Support via reports.</li></ul> |
| <b>Business Requirements:</b>                                | <ul style="list-style-type: none"><li>1- Update System</li><li>2- Provide On-line Access</li><li>3- IT Unit</li><li>4- Electronic Payment Gateway</li><li>5- Customer and Flight Data Management</li></ul>                                      |
| <b>Business value:</b>                                       | <ul style="list-style-type: none"><li>1- Saving time and effort</li><li>2- Increase Revenue and Sales</li><li>3- Reducing Operational Costs</li><li>4- Enhancing Customer Satisfaction</li></ul>                                                |
| <b>Special Issues and or Constraints:</b>                    | <ul style="list-style-type: none"><li>1- Deadline in 1 January.</li><li>2- Technical Challenges and Data Synchronization</li><li>3- Compliance and Security Standards</li></ul>                                                                 |

# Feasibility Analysis

## Technical Feasibility:

- 1- We need to design a responsive web interface using HTML5, CSS3, and JavaScript to ensure smooth user navigation across desktops and mobile devices.
- 2- We need development platforms and tools like visual studio code, MySQL Workbench, and Git/GitHub for version control and collaborative development.
- 3- We need a centralized database management system (such as MySQL or PostgreSQL) to manage flight schedules, passenger details, and real-time seat reservations.
- 4- We need to integrate payment gateway simulation and secure API connections to allow online ticket booking and automated confirmation.

## Economic Feasibility:

- 1- Initial development and setup costs are minimal, as the project utilizes open-source technologies, local development servers, and cloud-hosted databases.
- 2- The system reduces operational costs significantly by transitioning from manual paper-based ticketing to automated digital tickets (E-Tickets).
- 3- Revenue is generated through ticket booking commissions, seat selection preferences, and additional service fees.
- 4- The project offers a high Return on Investment (ROI) with a short payback period due to 24/7 automated booking capability without needing continuous human intervention.

## Organizational Feasibility:

- 1- **Management and Technical Capability:** The development team possesses the necessary skills in database administration, software engineering, and web development to deliver the system.
- 2- **Resource Sufficiency:** Essential non-financial resources, including hardware infrastructure, technical documentation, clear admin roles for travel agents, and system support guides, are fully available to ensure smooth adoption.